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The brain's internal echo: Longer timescales, stronger recurrent connections and higher neural excitation in self regions

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ABSTRACT

Background: Understanding the brain's intrinsic architecture has long been a central focus of neuroscience, with recent advances shedding light on its topographic organization along uni and transmodal regions. How the brain's global uni-transmodal topography relates to psychological features like our sense of self remains yet unclear, though.

Method: We here combine fMRI brain imaging with computational modeling (Wilson Cowan model) to better understand the temporal, spatial and physiological features underlying the distinction of self and non-self regions within the brain's global topography.

Results: fMRI resting state shows lower myelin content, longer timescales (measured by the autocorrelation window/ACW), and lower global functional connectivity/synchronization (measured by global signal correlation/GSCORR) in self regions (based on the three-layer self topography; Qin et al. 2020) compared to non-self regions. Next, we fit the fMRI data with a neural mass model, the Wilson-Cowan model, which is enriched by structural and functional connectivity data from human MRI/fMRI. We first replicate the empirical data with longer ACW and lower GSCORR in self regions. Next, we demonstrate that self and non-self regions can, based on the same measures in the model, not only be distinguished within the brain's global topography but also within the unimodal and transmodal regions themselves, respectively. Finally, the neural mass model shows that such topographic differentiation relates to two physiological features: self regions exhibit higher intra-regional excitatory recurrent connection and higher levels in their basal neural excitation than non-self regions.

Conclusion: Our findings demonstrate the intrinsic nature of the distinction of self and non-self regions within the brain's global uni-transmodal topography as well as their underlying physiological differences with higher levels in both recurrent connections and neural excitation in self regions. The increased recurrent connections in self regions, together with their higher levels of neural excitation and the longer autocorrelation window, may be ideally suited to mediate their self-referential processing: this can thus be seen as a form of 'psychological recurrence' where one and the same input/stimulus is processed in a prolonged echo-chamber like way, that is, an internal echo within the self regions themselves.

1. Introduction

1.1. How are self and non-self regions distinguished within the brain's global topography?

The concept of 'self' is a pivotal subject of investigation across psychology, psychiatry, and neuroscience, serving as a cornerstone in our understanding of human mental processes (Northoff, 2016; Qin et al., 2020; Qin and Northoff, 2011; Davey et al., 2019; Wolff et al., 2019) . The self is a core part of our mental life, encompassing sensory, motor, affective, and cognitive domains (Qin et al., 2020). Various models of the self have been previously hypothesized based on these

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domains, including the mental or cognitive or narrative self, the outer bodily self (Blanke et al., 2015), and the inner interoceptive self (Damasio, 2010; Seth and Tsakiris, 2018). Correspondingly, various neural correlates such as the cortical midline structures (Qin and Northoff, 2011), the temporoparietal junction (Blanke et al., 2015), and the insula (Qin et al., 2020) have been suggested. However, how these 'self regions' are distinguished from the rest of the brain, e.g., the 'non-self regions,' remains yet unclear. Interestingly, various studies show close relationship of the self with the brain's resting state activity (Qin and Northoff, 2011; Wolff et al., 2019; Davey et al., 2016; D'Argembeau et al., 2005; Huang et al., 2016; Kolvoort et al., 2020). Is the self and its distinction from the non-self thus an intrinsic feature of our brain's spontaneous activity? The goal of our study is to address this question by investigating whether and how the distinction of self and non-self regions is realized through and within the brain's intrinsic topography.

How can the brain distinguish self and non-self regions? The immune system's ability to distinguish between self and non-self antigens, producing antibodies against foreign entities while sparing the body's own cells, exemplifies a self/non-self discrimination at the molecular level (González et al., 2011). Similar to the immune system, neurons also must distinguish between internal and external stimuli, a process that is essential for the brain's interpretation and response to environmental cues (Cohen et al., 2022). Neurons respond selectively to external inputs while, at the same time, maintaining the integrity and stability of the brain's internal environment (Wainio-Theberge et al., 2022; Wainio-Theberge et al., 2021). This selective response helps in forming accurate perceptions and appropriate reactions to external stimuli.

The immune system has a clear mechanism in place for distinguishing self and non-self in its antigens and antibodies. In contrast to the immune system, we currently lack an understanding of the features through which the brain distinguishes regions related to the self and those associated with the non-self. Specifically, the structural and functional features of the brain's neural activity that facilitate their topographic differentiation across the brain as whole remain yet unidentified. Given that the self is associated with the brain's spontaneous activity (Qin and Northoff, 2011; Wolff et al., 2019; Davey et al., 2016; Huang et al., 2016; Kolvoort et al., 2020; Wolman et al., 2023; Wolman et al., 2023; Whitfield-Gabrieli et al., 2011; Bai et al., 2016; Smith et al., 2022), one might assume that the distinction of self and non-self is already manifest in the spontaneous activity – it may thus be an intrinsic features that is in-build within the brain's global topographic design and its underlying physiological features. Therefore, the goal of our study was to investigate those structural, functional, and physiological features in the brain's spontaneous activity and its topography, as measured during resting state activity, that allow for the differentiation of regions related to self from those associated with the non-self.

Are there specific regions in the brain uniquely related to the self? (Gillihan and Farah, 2005; Legrand and Ruby, 2009) While the debate remains yet to be settled, research confirms that certain core regions are associated with different aspects of the self. For instance, the mental or cognitive self has been linked to cortical midline structures (Araujo et al., 2013), while the bodily self is more connected with the temporoparietal junction (Eddy, 2016). Moreover, the interoceptive self, which processes the body's internal sensations, is associated with typical interoceptive regions like the insula and the thalamus (Berntson and Khalsa, 2021). A recent large-scale meta-analysis summarized these findings, suggesting a three-layer hierarchy of self in relation to regions involved with interoceptive, exteroceptive, and mental aspects of self (Qin et al., 2020). How these self regions are distinguished from non-self regions in the brain's topography through the latter's structural, functional and physiological features remains yet unclear, though. Addressing this question is the goal of our study.

1.2. Distinguishing self and non-self regions within the brain's global topography: myelin, autocorrelation window, global signal and their underlying physiological features

The exact structural, functional, and physiological features of this three-layer hierarchy of self regions, including its distinction from non-self regions, remain yet unclear though. One notable structural feature is the myelin content of neurons. Intracortical myelin content, estimated by the T1w/T2w ratio, has been widely used in research (Ito et al., 2020; Gao et al., 2020; Glasser and Van Essen, 2011). Distinct patterns of myelination have been observed across the cortex; unimodal areas such as the sensorimotor and primary visual cortices display elevated levels of myelination, while transmodal regions, including the prefrontal and cingulate cortices, exhibit lower myelin content (Nieuwenhuys, 2013; Glasser et al., 2013; Sui et al., 2022). Given that the self regions in the three-layer hierarchy include mainly higher-order or transmodal regions, one might expect the distinction between self and non-self regions to follow the uni-transmodal gradient of myelin thickness, with self-regions showing lower myelination than non-self regions.

Myelin also relates to functional features like the brain's intrinsic neural timescales (INT). Cortical myelin thickness has been established as a non-invasive marker for anatomical hierarchy: highly myelinated regions such as sensory cortices exhibit shorter intrinsic timescales (INTs), while less myelinated areas, like the transmodal association cortices, display longer timescales (Ito et al., 2020; Gao et al., 2020; Wang, 2020). Furthermore, there is evidence that recurrent excitatory activity mediated by NMDA receptors increases progressively from sensory regions to higher-order association cortices (Gao et al., 2020; Wang, 2020). Computationally, it has been shown that increased excitatory activity causes long-range autocorrelations of those neural firings (Deco et al., 2014). Together, these findings suggest that myelination, as a proxy for cortical hierarchy, can offer valuable insights into the balance of excitation and inhibition (E-I) across the cortex, particularly through its influence on recurrent excitatory activity and basal excitation dynamics (Kong et al., 2021; Zhang et al., 2023).

The autocorrelation window (ACW) serves as a measure of intrinsic neural timescales, capturing the temporal persistence of neural activity within specific brain regions. Regions with lower myelin content demonstrate longer autocorrelation windows (ACW) as a measure of INT, which can be explained by a negative correlation between myelin content and INT (Ito et al., 2020). Notably, the mental or cognitive self has been associated with longer ACW (Wolff et al., 2019; Huang et al., 2016; Kolvoort et al., 2020; Wolman et al., 2023; Smith et al., 2022). Given that the ACW follows the myelin gradient from sensory unimodal to associative transmodal regions explained as above), one might assume that typical self regions like cortical midline regions exhibit longer ACW than non-self regions.

Another distinguishing feature between unimodal and transmodal regions is the global brain activity (global signal/GS) and its topography as measured by global signal correlation (GSCORR). GSCORR reveals the spatial distribution of the GS and provides insights into how different brain regions correlate and thus synchronize with global brain activity: higher GSCORR has been observed in unimodal sensory regions than in higher-order transmodal association cortices (Zhang et al., 2020; Yang et al., 2014; Zhang and Northoff, 2022). How the topography of GSCORR relates to the distinction of self- and non-self regions is not yet well understood, though.

1.3. Aims – structural, functional and physiological differentiation of self and non-self regions within the brain's uni-transmodal topography

First aim – Using fMRI resting state to investigate myelin, timescales and global signal in self and non-self regions: Our first aim is to distinguish self and non-self regions in resting state fMRI based on the three-layer hierarchy of self (Qin et al., 2020), by focusing on their structural (e.g., myelin content), functional-temporal (e.g., INT with ACW), and functional-spatial (e.g., GSCORR) features. As noted earlier, self-related regions exhibit longer ACW and lower GSCORR in transmodal areas compared to unimodal ones. Given the known negative relationship between INT and myelin (Ito et al., 2020), we hypothesize that self-related regions will have lower myelin content, longer ACW and lower GSCORR compared to non-self regions. To test this hypothesis, we will apply standard measures to a large-scale fMRI dataset from the Human Connectome Project (HCP). We also utilized the transmodal-unimodal atlas (Ito et al., 2020) that intersects with self-nonself regions to divide regions into four groups: transmodal-self (TS), transmodal-nonself (TNS), unimodal-self (US), and unimodal-nonself (UNS). This categorization allowed us to test whether the differences between self and non-self regions are intrinsic by themselves (that is, distinct from the uni-transmodal distinction itself) rather than just being a byproduct of the transmodal-unimodal differentiation. Additionally, we applied bootstrapping to all regions classified as self or nonself to assess whether the observed differences could be attributed to chance. We also controlled for preprocessing effects, such as bandpassing and global signal regression, to illustrate how these steps influence the differences between categories (see Supplementary Material).

Second aim – Relationship among myelin, timescales, and global signal in self and non-self regions: Our second specific aim is to investigate the relationships between the three measures—myelin content, ACW, and GSCORR—both globally within the whole brain with its uni-transmodal topography and specifically within self and non-self regions. Based on previous findings mentioned above, we anticipate that myelin content will be negatively correlated with ACW and positively correlated with GSCORR in both self and non-self regions. This hypothesis will be explored through a Bayesian Mediation Model.

Intrinsic neural timescales (INT) are crucial for organizing functional connectivity (FC) throughout the brain. Recent studies using ultrahigh field fMRI in primates have shown that these timescales are arranged along spatial gradients that correspond to FC gradients, underscoring their importance in large-scale network organization. Notably, longer INTs exhibit unique connectivity profiles (Manea et al., 2022). Variations in INTs impact functional outcome of the cortex within both sensory and cognitive domains—slower timescales are linked to cognitive involvement (Zeraati et al., 2023), whereas faster timescales enhance connectivity to visually responsive areas and claimed that FC and INT influence one another (Boring et al., 2022) .

Further, it has been demonstrated that INT's change the FC of working memory (Wasmuht et al., 2018). Additionally, disruptions in INTs, whether through pharmacological sedation or cortical inactivation (Huang et al., 2018; Soyuhos et al., 2024), result in altered FC and cognitive deficits. These findings highlight how INTs dynamically regulate FC, influencing both spontaneous and task-related brain activity. We therefore assumed close relationship of INT/ACW with the GSCORR as the latter measures the brain's global FC; however, we assumed that this relationship will be slightly different for self and non-self regions.

Our conceptual model of the relationship of myelin, ACW and GSCORR is grounded in prior work showing that local cortical properties (e.g., myelin content, which we treat as a proxy for anatomical hierarchy and recurrent excitation) shape the Intrinsic neural timescales (INTs) of the local network (Wang, 2020; Chaudhuri et al., 2015). ACW indexes local timescales by capturing intra-regional neural signal correlations. Global Signal Correlation (GSCORR), by contrast, reflects large-scale synchronization across the cortex, meaning an inter-regional measure.

Together, these different lines of evidence, suggest that ACW sits naturally "in between" a structural/physiological measure (myelin) and a global functional measure (GSCORR). These considerations motivated us to treat ACW as the mediator of the relationship between myelin and GSCORR. In principle, one could also test an alternative model—e.g., GSCORR mediating the myelin–ACW relationship—but such a model would be less consistent with current theories that local physiological properties (myelination, excitation) drive local intra-regional neural timescales, which, in turn, scale into the brain's global inter-regional dynamics. We will test these different models using Bayesian statistics.

Third aim – Using computational modelling to detect the underlying physiological features of self and non-self regions: Building on the evidence that intrinsic neural timescales (INTs) and functional connectivity (FC) are influenced by the balance of excitation and inhibition across cortical hierarchies, our third specific aim focuses on understanding how basal excitatory tonus and recurrent excitatory connections within the brain regions mediate the transition from the structural properties of myelin to the functional dynamics of the autocorrelation window (ACW) and global signal correlation (GSCORR). Previous computational studies demonstrated that both basal excitatory activity and recurrent excitatory connections are crucial for local-to-global coupling and the regulation of local oscillations (Deco et al., 2014; Kong et al., 2021; Papadopoulos et al., 2020). Therefore, we aim to explore how these two forms of neural excitatory dynamics serve as a bridge between the anatomical hierarchy (myelin content) on the one hand and the functional outcomes like ACW and GSCORR on the other and, how that, allows distinguishing self and non-self regions.

To achieve this, we will apply computational modeling using the Wilson-Cowan neural mass model enriched by the structural and functional connectivity data of human MRI/fMRI; after fitting the data to the model, this allows us to simulate and investigate the interplay of physiological features like global coupling, local basal excitatory tone, recurrent excitatory connections with the neuronal measures of myelin content, ACW, and GSCORR. That served the purpose to distinguish self and non-self regions through their underlying physiological features like recurrent connections and neural excitation. We also examined whether local dynamical parameters (e.g., myelin and excitatory recurrent connections) are more strongly associated with an intraregional metric such as ACW, whereas an intra-node parameter—like global coupling—is more closely related to global signal correlation (GSCORR).

Given that both recurrent connections and neural excitation also exhibit differences along the uni-transmodal gradient (see above), we hypothesized that self regions show higher degrees in both their recurrent excitatory connection and basal excitatory tone than non-self regions. Together, higher levels of neural excitation and higher recurrent connections, indicative of self-regions at the neuronal level, could provide a neurobiological basis for self-referential processing that may potentially reflect a form of psychological recurrence. The increased recurrent connections, together with higher levels of neural excitation, may thus provide the physiological basis for the increased recurrent processing of one and the same input or stimulus within the self regions which, on the neural level, is facilitated by the longer ACW while psychologically it may be manifest in self-referential processing as form of "psychological recurrence" through the prolonged recursive processing of one and the same input/stimulus like an internal echo within the self regions.

2. Results

2.1. *fMRI resting state I: self regions show lower myelin content, longer ACW and higher GSCORR than non-self regions*

We first investigated whether self and non-self regions show different degrees of myelin as well as differences in their ACW and GSCORR. ACW was calculated as the lag where autocorrelation function reaches 0 whereas GSCORR was defined as the pearson correlation between the BOLD activity of a region and the averaged BOLD activity of the gray matter across cortex. Myelination values were estimated as T1w/T2w ratio of structural MRI (See Supplementary Material for further details). When averaging regions across subjects (We call as 'subject-wise'), we observed that self regions have significantly less intracranial myelin (MY) content compared to non-self regions ($t(163) = 62$, $p<.001$, $d = 4.9$, Fig. 1A-left). The autocorrelation window (ACW) was significantly longer in self than non-self regions ($t(163) = 10.2$, $p<.001$, $d = 0.795$,

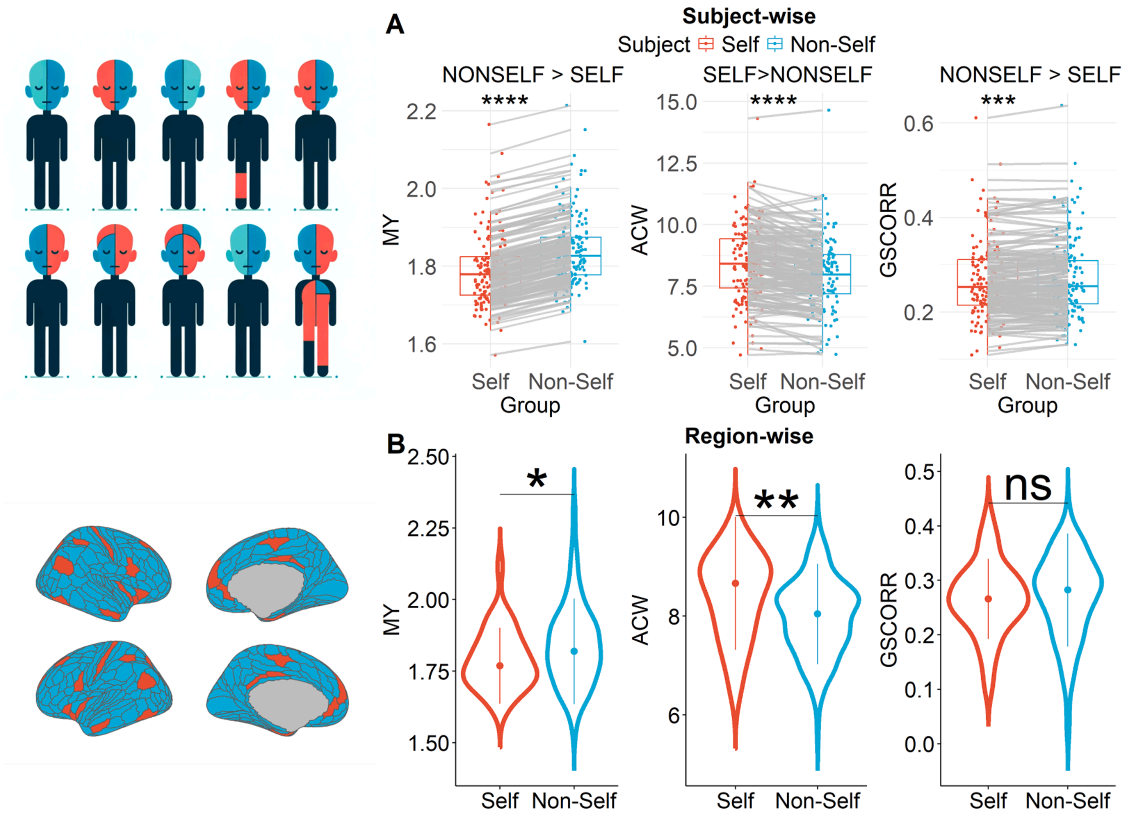

Fig. 1. Comparison between Self and Non-Self regions: A) Comparative Analysis of Intracranial Myelin Content, ACW, and GSCORR Between Self and Non-self Regions as based on the three layer hierarchy of self (Qin et al. 2020). Cartoon showing subject-wise analysis by averaging regions within categories per subject. Left: Myelin content is significantly lower in self regions compared to non-self regions. Middle: Self regions show significantly longer ACW compared to non-self regions. Right: GSCORR is notably higher in non-self regions than in self regions. B) The brain shows self (red) and non-self (blue) regions, highlighting that B is a region-wise analysis averaged across subjects. When averaging data for regions, self regions again exhibit significantly lower intracranial myelin and longer compared to non-self regions. The difference in global signal correlation (GSCORR) was not significant in this region based approach, though.

Fig. 1A-middle). Additionally, non-self regions exhibit higher GSCORR than self regions. ($t(163) = 3.66$, $p<.001$, $d = 0.286$, Fig. 1A-right)

In addition to averaging across subjects, we also averaged for regions rather than subjects (called as 'subject-wise'). The Shapiro-Wilk test suggests that the self variable follows a normal distribution, whereas non-self GSCORR and MY may not. Consequently, the Mann-Whitney U test was employed for MY and GSCORR, while the Student's *t*-test was used for ACW. Consistent with the previous analysis, less intracranial myelin content is observed in self regions compared to non-self ($W = 4251$, $p<.05$, $r = 0.106$, Fig. 1B-left). ACW was significantly longer in self than in non self regions ACW ($t(37) = 2.82$, $p<.05$, $r = 0.53$, Fig. 1B-middle). However, in this case, GSCORR was not significantly different between self and non-self regions ($W = 4949$, $p=.43$, $r = 0.041$, Fig. 1B-right). In sum, self regions show lower myelin content as well as longer ACW and lower GSCORR than non-self regions.

We performed comparisons under several preprocessing conditions, specifically bandpassing between 0.01 Hz up to the Nyquist frequency, both with and without global signal regression (GSR). Across all preprocessing variations, we observed that the autocorrelation window (ACW) values were higher in self-related regions compared to non-self regions (Supplementary Table 3 and 12). Similarly, global signal correlation (GSCORR) values were higher in non-self regions, when global signal was included, with and without bandpassing (Supplementary Table 15). These findings indicate that the observed group differences are robust for to the preprocessing methods employed in subject-wise analysis(see Supplementary Material for a detailed explanation).

To control for potential confounds in our analysis, we examined whether the self/non-self distinction merely reflects the unimodal/transmodal distinction, with unimodal regions corresponding to non-self and transmodal regions corresponding to self. Our findings indicate that, across all preprocessing combinations, the autocorrelation window (ACW) is higher in transmodal regions compared to unimodal regions. Conversely, global signal correlation (GSCORR) is higher in unimodal regions compared to transmodal regions, except when GSR is applied. As a control for GSCORR, we also calculated global functional connectivity (GFC), where correlation coefficients are averaged, instead of BOLD signal (See Supplementary Material – Global Functional Connectivity). Global functional connectivity (GFC) showed a similar pattern to GSCORR across all preprocessing combinations (See Supplementary Material).

We then further divided the transmodal and unimodal regions into self and non-self regions. The results consistently demonstrate a reverse hierarchy, with transmodal self regions exhibiting the longest ACW, descending to unimodal non-self regions with the shortest ACW (See Fig. 4B, also Supplementary – Table 37). For GSCORR and GFC, the pattern is reversed (see Supplementary Material/Self-Modality comparison for detailed results). Results indicates that self and non-self distinction is independent from unimodal/transmodal distinction, as having their respective places within unimodal and transmodal.

Additional to that, we bootstrapped the labels of self and non-self regions to determine if the observed difference was due to chance. Our findings, detailed in the Supplementary Material - Bootstrapping, reveal a noteworthy distinction in the significance levels across different parameters when our bootstrapping methodologies are applied. Specifically, the empirical differences between ACW and myelin content is statistically significant in both random and controlled shuffling (threshold for p-value is 0.05, See Supplementary Figure S1 and S2). This underscores the importance role of myelin and ACW in distinguishing self-related regions in our data sample and differences disappear when data is shuffled. Lastly, we validated the subject-wise differences between self/non-self regions with topography using the data from the 3T S1200 dataset, focusing on a group of 100 unrelated subjects (Figure S9 and S10).

2.2. fMRI resting state II: relationship of myelin with ACW and GSCORR

Given the well known relationship of myelin and ACW (Ito et al., 2020), we correlated the myelin with both ACW and GSCORR as well as the ACW with GSCORR in a global way across all regions in the brain. As expected, ACW and myelin content showed strong negative correlation across regions (Spearman correlation, $\rho = -0.45$, $p_{spin} < 0.001$, Fig. 2A

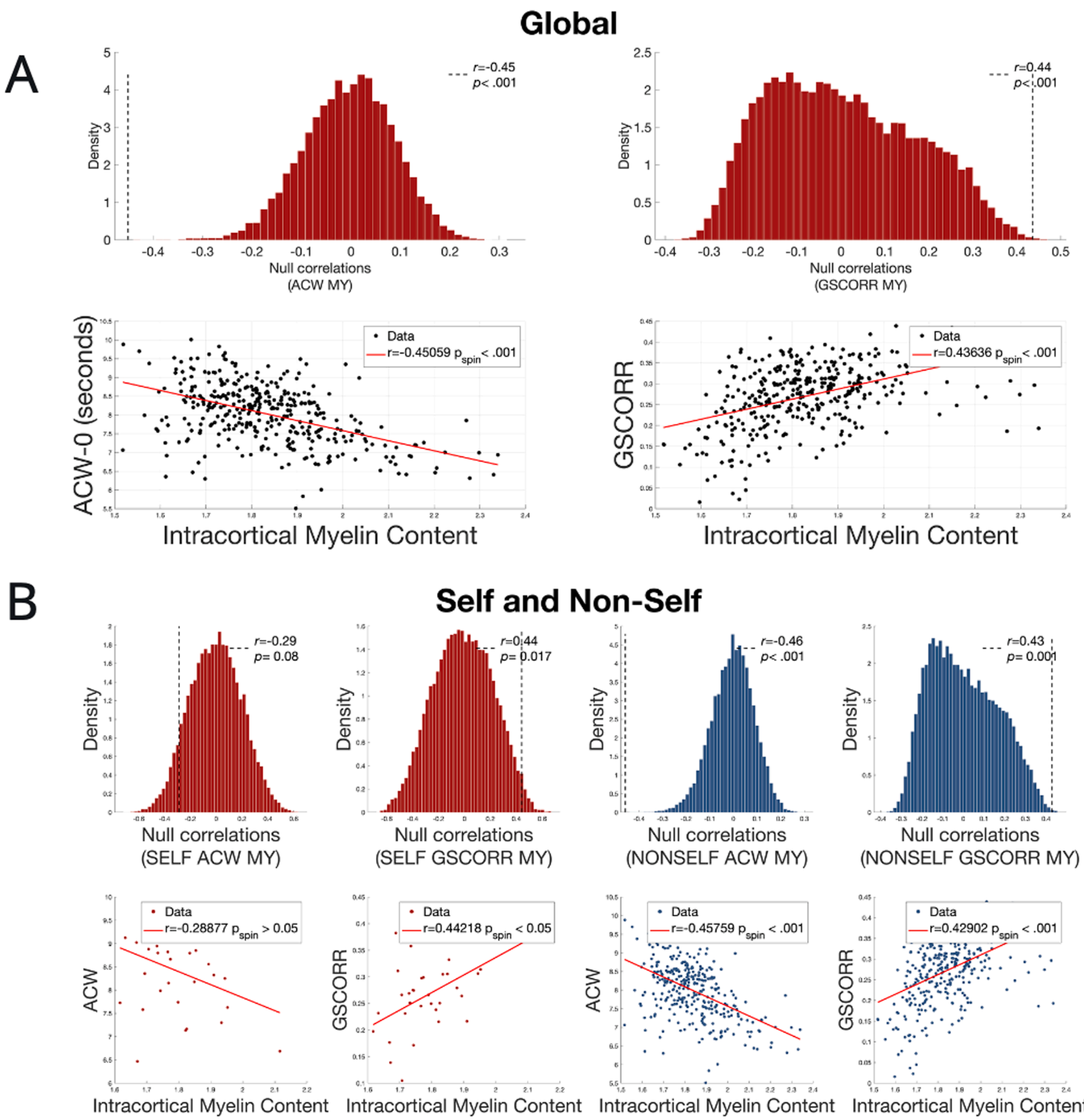

Fig. 2. Correlation Analysis of ACW, GSCORR, and Myelin Content Across All Brain Regions: A) On the left, a strong negative correlation between ACW and myelin content is depicted (Spearman's rho = -0.45, $p_{spin} < 0.001$), ascertained using a spin permutation test for statistical significance. On the right, GSCORR's positive correlation with myelin content (Spearman's rho = 0.44, $p_{spin} = 0.028$) after data shuffles 10 (Wolff et al., 2019) times. B) Different Correlations of Myelin Content with ACW and GSCORR in Self and Non-self Regions. This analysis dissects their relationships within self and non-self classified regions, highlighting a non-significant correlation between myelin content and ACW in self regions (rho=-0.29,$p_{spin}>0.05$), alongside a significant positive correlation between myelin content and GSCORR (rho= 0.44 , $p_{spin}<0.05$). While non-self regions exhibit a strong negative correlation between myelin content and ACW comparable to self and global correlations (rho = -0.46,$p_{spin}<0.001$). The correlation between GSCORR and myelin in non-self regions is significant and positive, mirroring the global pattern (rho=0.43, $p_{spin} < 0.001$).

left side), with significance (p) value defined by spin permutation test (See, Methods). GSCORR and myelin also correlated though in a positive way after 10 (Wolff et al., 2019) data shuffles (p_{spin}=0.001), with a rho coefficient of 0.44 (Fig. 2A right side).

When regions are divided to self and non-self for correlation analysis, self regions demonstrate no correlation between myelin content and ACW ($\rho = -0.29$, $p_{spin} > 0.05$), along with strong positive correlation of myelin content with GSCORR ($\rho = 0.44$, $p_{spin} < 0.05$, Fig. 3 left side). Non-self regions also demonstrate strong negative correlation between myelin content and ACW ($\rho = -0.46$, $p_{spin} < 0.001$) with GSCORR positively related to myelin content ($\rho = 0.43$, $p_{spin} < 0.05$).

In summary, the results show strong global negative relationship of myelin content with the ACW while its relationship with GSCORR is positive. This global pattern is especially pronounced in the non-self regions and no significant correlation between ACW and MY in self regions.

Finally, we investigated whether the observed difference was influenced by the number of regions. To control for this, we selected non-self regions to match the number of self regions using k-means clustering to balance the dataset, we used k-means clustering on the majority class (nonself, $k = 5$) and randomly selected samples from each cluster in proportion to the minority class size (self). If a cluster had more samples than needed, a random subset was taken; otherwise, all samples were included. We adjusted the sample count to match the minority class by

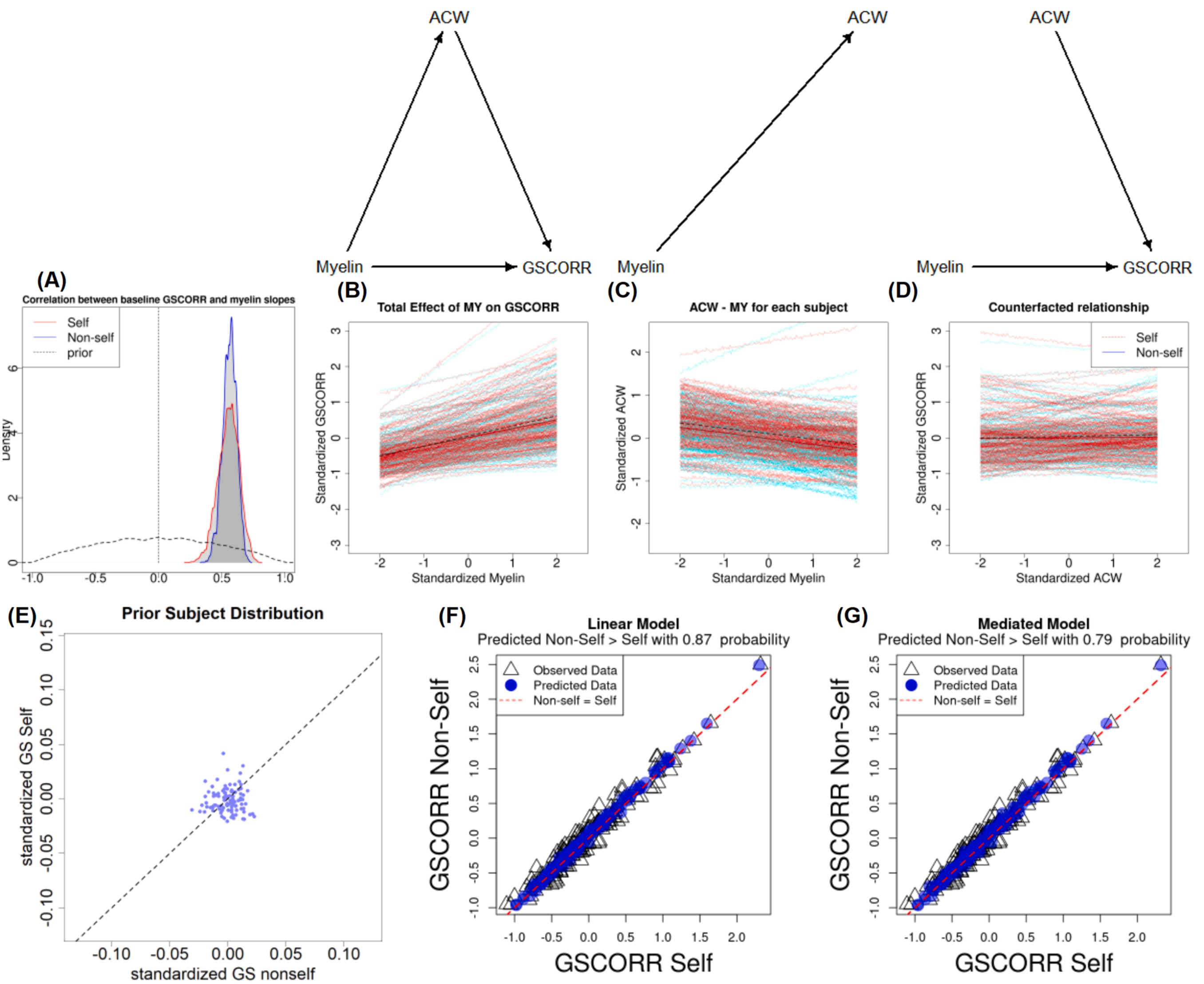

Fig. 3. Hierarchical Varying Intercept and Slopes Model: To avoid averaging data in a way that loses information about uncertainty, we conducted a varying effect model with subjects. Regions are categorized as self and non-self. We employed a prior multivariate Gaussian distribution across estimates to investigate potential correlations between them. A) The posterior distribution of rho values between baseline GSCORR and slopes of myelin indicates that higher baseline GSCORR results are associated with higher myelin content across subjects) The posterior distribution of rho values between baseline GSCORR and slopes of myelin indicates that higher baseline GSCORR results are associated with higher myelin content across subjects. C) As part of a counterfactual analysis, when myelin is controlled from −2 SD to +2 SD, the model captured a group average slope of −0.13 (CI: [−0.29, 0.12]) in non-self regions and −0.12 (CI: [−0.26, 0.03]) in self regions, reflecting uncertainty in relationship between ACW and MY with both positive and negative for non-self while being mostly negative in for self regions.D) When the path from myelin to ACW is disrupted in silico, the slopes become uncertain (Self: 0.02, CI: [−0.23, 0.27]; Non-self: 0.01, CI: [−0.23, 0.27]), reflecting the mediation role of ACW in this interaction. The results show that, when intersubjective variability is included in the model (compared to the model that did not include it in Figure 2), both self and non-self regions exhibit full mediation of ACW. E) The model's prior assumptions show that GSCORR values for both self and non-self regions are initially distributed around 0, without tilting or bias. F) After the Bayesian procedure with the data, the model shows an increased GSCORR for non-self regions with an 87 % probability in the linear model. Each dot represents a simulated subject sampled from the posterior distribution. G) The ACW mediation model shows an increased GSCORR for non-self regions with a 79 % probability in simulated subjects from the posterior distribution.

trimming or adding samples. Finally, the undersampled majority and minority class samples were combined to create a balanced dataset, with labels reflecting the new distribution. The relationship remained unchanged after adjusting for the number of regions (see Supplementary Material - Controlling number of regions for details, Figure S3 and S4).

2.3. fMRI resting state III: exploring the relationship between myelin, ACW, and gscorr - using hierarchical bayesian modeling

To account for intersubjective variability in the data, we applied a hierarchical multi-level Bayesian model across subjects (for details, see Methods – Hierarchical Multi-Level Bayesian Model). This model enables us to pool effects across subjects and regions, thereby regularizing the estimates. We employed Multivariate Gaussian distribution as a prior across estimates to investigate potential correlations between them. Two models were used: one is only including myelin (MY) and GSCORR (linear model), and one alternative model with ACW as a mediator from (mediated model) from myelin to GSCORR. Markov Chain Monte Carlo (MCHC) sampling was effective and chains converged (Supplementary Material - MCHC Convergence Diagnostics, Figure S5 and S6). A posterior predictive check demonstrated that the model captured the data distribution (Supplementary Material - Posterior Predictive Checks, Figure S7 and S8).

In the mediated model, the posterior distribution of ρ values between baseline GSCORR and slopes of myelin indicates that higher baseline GSCORR results are associated with higher myelin content across subjects (see Fig. 3A). This pattern is consistent across for self and non-self regions ($\rho_{self} = 0.54$ (CI: [0.41, 0.68], $\rho_{non-self} = 0.55$ (CI: [0.47, 0.65])) These results suggest that the effect of structural substance (myelin) on functional outcome (GSCORR) is more pronounced, with higher baseline in myeline results in higher GSCORR.

To demonstrate the mediation of ACW for self and non-self regions for each subject, we performed a counterfactual analysis (see Methods, Counterfactual Analysis) (McElreath, 2020). This analysis plots controlled parameters against learned parameters across subjects, reflecting the outcome of interest and how it changes according to perturbation. When myelin is controlled from -2 SD to $+2$ SD, similar to Figures 2, the total effect of myelin on GSCORR shows an average slope of interaction of 0.28 with intersubjective variability [CI: 0.07, 0.51] in self regions, while it is 0.24 [CI: 0.01, 0.45] in non-self regions (Fig. 3B). It important to highlight that the model captured the positive association between MY and GSCORR in consistent with figure 2.

Similarly, we show that the model consistently captures a negative relationship between myelin and ACW, with a group average slope of -0.13 (CI: [-0.29, 0.12]) in non-self regions and a slope of -0.12 (CI: [-0.26, 0.03]) in self regions, with varying intercepts on the Y-axis (Fig. 3C). The confidence intervals indicate an uncertainty in the myelin-ACW relationship in non-self regions but greater certainty in self regions. This suggests that, when subjects are not averaged per region and information is pooled across subjects, self regions demonstrate a clearer negative relationship between myelin and ACW.

To complete the counterfactual analysis, we disrupted the path from myelin to ACW in silico. The slopes become uncertain in both self and non-self (Self: 0.02, CI: [-0.23, 0.27]; Non-self: 0.01, CI: [-0.23, 0.27]) – this reflects the role of ACW as mediator (Fig. 3D). These results suggest that, when intersubjective variability is included in the model (compared to a model that did not include it, as in Figure 2), both self and non-self regions show mediation effect of ACW on the relation from myelin to GSCORR.

We also investigated the posterior distribution of GSCORR as learned by the model. In Fig. 3E, the model's prior assumptions show that GSCORR for both self and non-self regions are distributed around 0, without tilting or bias. After applying the Bayesian procedure with the data, the linear (87 %) and mediated (79 %) models resulted in increased GSCORR for non-self regions, reflecting the effect seen in Fig. 1A. This suggests that GSCORR is higher in non-self than in self, which is learned by the model. Posterior predictive checks for GSCORR are shown in our supplementary material for ten random subjects, their posterior distributions successfully reproduce the observed data's patterns and variability.

After demonstrating the significant impact of ACW on the relationship between myelin and GSCORR, we showed that the predictive accuracy of the mediation model surpasses that of the linear model, as measured by Pareto Smoothed Importance Sampling (PSIS) (see Supplementary Material - Comparing Linear and Mediated Models). It is important to note that the PSIS penalizes including new parameters to prevent overfitting. Despite this, the predictive accuracy still favors the mediation model (see Supplementary Material – Table S42). Additionally, we tested the potential mediation effect of GSCORR. However, it showed significantly lower performance for both mediated and linear models (see Supplementary Material – Table S42).

In summary, both self and non-self regions show mediation from myelin to GSCORR; however, uncertainty is higher in the myelin-ACW relationship for non-self regions. This underscores the importance of pooling information across subjects and considering intersubject variability, instead of averaging when comparing ACW-myelin results with Figure 2.

2.4. Computational modeling I: simulating hierarchical differences in neural activity – dependence of ACW and gscorr on neural excitation and recurrent connection

Upon demonstrating the statistical dependence of GSCORR on myelin content and ACW, we utilized a biophysical model to replicate the empirical observations (see Methods - Neural Mass Modelling and BOLD Transformation). This approach was used to test the hypothesis that myelin-based differences in ACW and GSCORR are computationally related to excitatory recurrent connections and excitatory activity. We found that myelin content increases from transmodal self to unimodal non-self regions, while ACW shows an inverse relationship. For GSCORR, the transmodal self regions exhibited higher values than the transmodal non-self regions, with no significant difference between unimodal self and unimodal non-self regions. While in total, unimodal regions generally displayed higher GSCORR than transmodal regions (Fig. 4B). For contrast tables between groups, see Supplementary Material Self-Modality comparison (Table S38–41).

We tested the hypothesis that the observed topographical differences in ACW and GSCORR could depend on recurrent excitatory strength and excitatory tone, which are scaled according to hierarchical levels, with myelin serving as a proxy. Previous research has shown that the excitation-to-inhibition balance (EIB balance) increases from unimodal to transmodal areas (Fotiadis et al., 2023). Similarly, we propose that recurrent excitatory strength and excitatory tone may vary with hierarchical position according to myelin, contributing to an increased excitatory rate at higher hierarchical levels and contributes self/non-self difference with uni/transmodal.

To evaluate our hypotheses, we conducted simulations using a Wilson–Cowan neural mass model. First, we performed a single-node simulation, systematically excitatory recurrent connections (see Methods). Results showed that neural fluctuations emerged within the recurrent connection range of 15–20.8. Within this range, the autocorrelation window (ACW) increased monotonically with recurrent connectivity (Spearman's correlation: $\rho = 0.612, p < .001$). This shows local recurrent connections to a local, i.e. intraregional, measure such as ACW. This finding supports us to assign more excitatory recurrent connections to the regions that have longer ACW, with myelin as proxy.

We explored three distinct scenarios: first, we systematically varied combinations of recurrent connections and excitatory activity, adjusting their means and variances according to myelin content, and selected parameter sets that matched empirical observations. In the second scenario, we fixed the recurrent connections to the values estimated in the

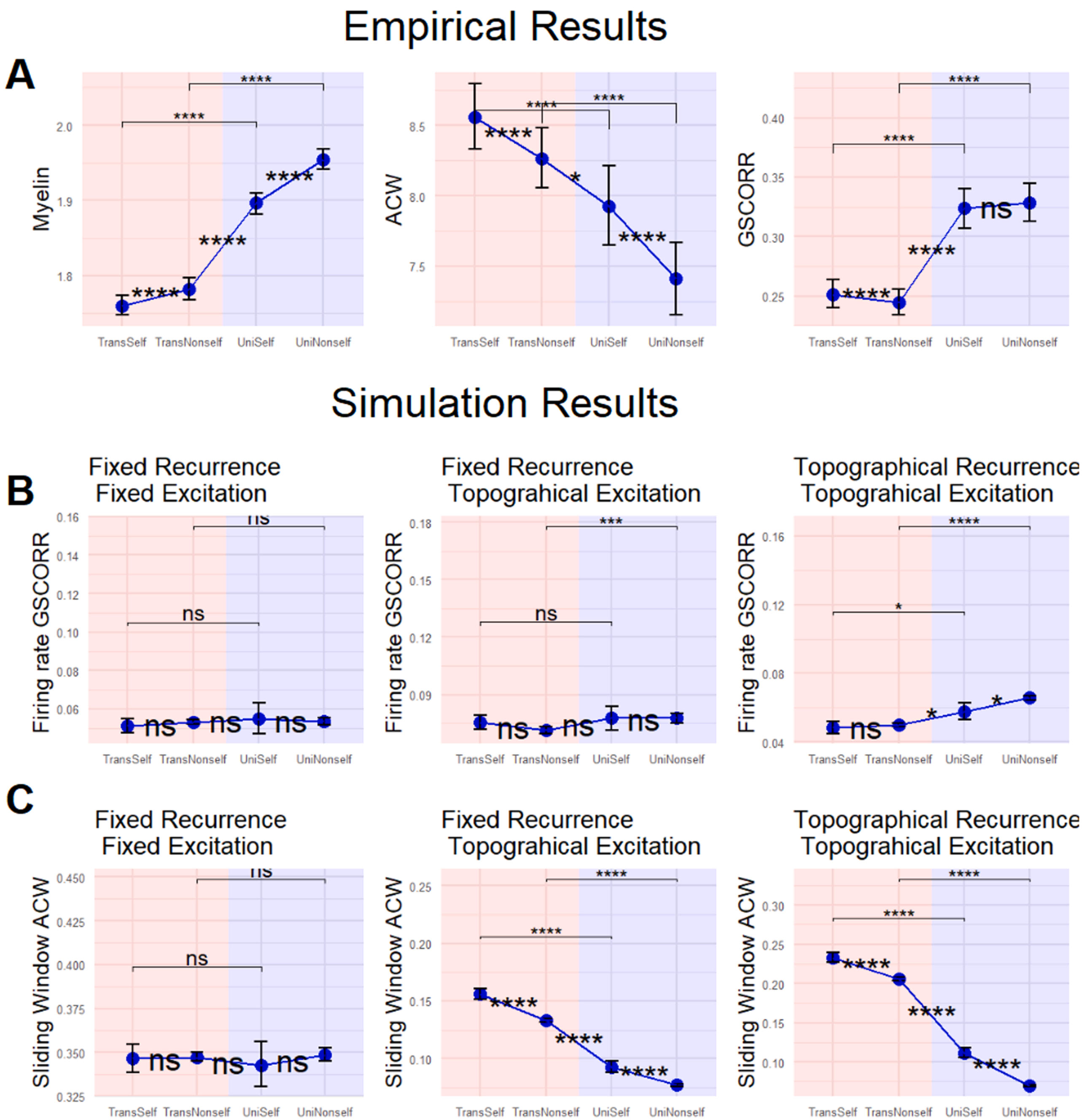

Fig. 4. Simulation with Wilson – Cowan Neural mass modelling and replication of empirical results. A) Empirical results of four categories. Myelin levels increased from transmodal self to unimodal nonself. Notably, unimodal areas exhibit higher myelin levels compared to transmodal areas, and nonself areas show higher myelin levels than self areas. The categories demonstrate a decrease in ACW from transmodal self to unimodal nonself, which reflects a negative correlation between ACW and Myelin, as illustrated in Figure 2 of the main text. Global Signal Correlation (GSCORR) displays a similar pattern to myelin, with higher values in unimodal areas compared to transmodal areas. However, when comparing self and nonself categories, GSCORR is higher in self areas, as depicted in Fig. 1. Global Functional Connectivity (GFC) shows a topography similar to GSCORR. This similarity indicates that GFC serves as a control for averaging time series or correlation coefficients, maintaining consistent patterns across different measures (Supplementary Material). B) Simulation results show the effects of different neural scaling approaches on the firing rate of GSCORR across the hierarchical brain regions. (Left panel) When nodes excitatory connections and excitatory baseline is fixed, without myelin as topographical information, there is no statistical difference between regions. (Middle panel) When nodes' excitatory tone are scaled according to myelin as a topographical proxy, only unimodal nonself is higher than unimodal self, in the firing rate GSCORR.(Right Panel) Scaling individual excitatory recurrent connections and excitatory tone inversely according to the myelin demonstrates increased GSCORR in unimodal regions compared to transmodal regions, with unimodal nonself regions showing higher GSCORR than unimodal self and transmodal nonself regions; this aligns, closely with the empirical observations. C) Simulation results showing the effects of different neural scaling approaches on ACW values across the hierarchical brain regions. (Left panel) Identical nodes simulation shows no difference among regions. (Middle panel) Only scaling the baseline excitation according to myelin as a topographic proxy results in replicated ACW similar to topographic difference. (Right panel) Inverse scaling of individual excitatory recurrent connections and baseline excitatory activity according to myelin replicates empirical findings, with clear differentiation between unimodal and transmodal regions with self and non-self regions, showing differences consistent with empirical data. Together, these findings demonstrate the key role of recurrent connection and excitatory tone in yielding the topographic-dynamic distinction of self and non-self regions across the unimodal-trans modal distinction within the brain's spontaneous activity.

first scenario but removed any variance associated with myelin to highlight the impact of myelin scaling on recurrent connectivity. In the third scenario, we kept the recurrent connections and basal excitatory activity constant, using the estimated mean values from the first scenario without applying any scaling related to myelin, to isolate the effect of maintaining fixed recurrent connections and excitation levels from any myelin-driven changes.

Firstly, we introduced excitatory recurrent connections and excitatory baseline activity, both scaled inversely according to myelin, which served as topographical information. We utilized a combined score for the best fit of the model to data (see Methods) . Parameter space exploration, using various means and standard deviations, among the best ten scores revealed that the greatest differences between regions—consistent with empirical findings—occur when the mean excitatory recurrent connectivity strength is set to 20 with a standard deviation of 2, and the baseline excitatory activity is set to 0.8 with a standard deviation of 0.2.

When we simulated empirical data with these parameters, the results revealed higher GSCORR in unimodal regions compared to transmodal regions. Specifically, unimodal non-self regions showed significantly higher GSCORR than unimodal self regions, while no significant differences were observed between transmodal regions. These results are consistent with the empirical data: TransSelf vs. UniSelf (estimate $= -0.00931$, $t(59) = -2.88$, $p < .05$, $d = -0.74$), TransSelf vs. UniNonself (estimate $= -0.017$, $t(59) = -9.09$, $p < .0001$, $d = -1.385$), TransNonself vs. UniSelf (estimate $= -0.007$, $t(59) = -3.15$, $p < .05$, $d = -0.616$), TransNonself vs. UniNonself (estimate $= -0.015$, $t(59) = -17.76$, $p < .0001$, $d = -1.258$), and UniSelf vs. UniNonself (estimate $= -0.00883$, $t(59) = -3.12$, $p < .05$, $d = -0.642$). Furthermore, ACW results indicated that ACW was longest in transmodal self regions, gradually shortening through unimodal non-self regions along the cortical hierarchy. (Fig. 4).

We simulated the baseline excitation of each node, scaling it inversely according to the myelin, while keeping the recurrent connections fixed as estimated above ('c_excexc' = 20). In terms of firing rate GSCORR, only unimodal nonself showed a higher GSCORR compared to transmodal nonself (TransNonself vs UniNonself, estimate $= -0.006$, $t(59) = -4.04$, $p = .001$, $d = -0.411$). For ACW, the results replicated the same topographical differences from transmodal self to unimodal nonself. This suggests that recurrent connections play an important role in the observed differences in the empirical data, particularly for the difference of ACW and GSCORR in self and non-self difference. Finally, we conducted simulations using identical nodes with a fixed excitatory recurrent parameter ('c_excexc' = 20) and a fixed basal excitatory activity ('c_ext_baseline' = 0.8). The results indicated no significant differences in GSCORR or ACW between regions. Overall, these simulation outcomes did not match the empirically observed patterns (see Fig. 4B and C, left side).

2.4.1. Computational modeling II: effects of local (recurrence, neural excitation) and global parameters (global coupling) on Intra- and inter-regional measures (ACW, GSCORR)

After demonstrating the importance of a topographical approach in replicating empirical findings, we investigated how our key model parameters influence two neural measures: autocorrelation window (ACW) and global signal correlation (GSCORR). Based on theoretical expectations, local excitatory parameters (i.e., excitatory recurrent connections and basal excitatory tone) should primarily modulate intraregional measures such as ACW, whereas interregional parameters (e.g., global coupling) should predominantly affect interregional connectivity measures such as GSCORR. To test this hypothesis, we applied a mixed-effects model (see Methods) to assess the contributions of these parameters.

Results showed that recurrent excitatory connections, after accounting for regional category (US, UNS, TS, TNS), explained a substantially larger proportion of variance in ACW (marginal $R^2 = 0.138$, conditional $R^2 = 0.469$) compared to GSCORR (marginal $R^2 = 0.043$, conditional $R^2 = 0.120$). In contrast, global coupling contributed more to GSCORR (marginal $R^2 = 0.065$) than to ACW (marginal $R^2 = 0.010$).

Higher marginal R2 values in recurrent connections to ACW and global coupling to GSCORR support the hypothesis that interregional parameters such as global coupling predominantly shape interregional functional connectivity (GSCORR), while local excitatory parameters more strongly influence local neural dynamics, e.g., ACW. This highlights the alignment between the measures in data and model, and the model where the two of them might map onto each other.

2.4.2. Computational modeling III: replication of the mediation analysis of fMRI resting state in the simulation data

Next, we examined whether the simulated data demonstrated the mediating role of ACW by fitting the hierarchical Bayesian model to the synthetic data. We found that recurrent connections were positively associated with ACW and negatively associated with GSCORR, with these effects being particularly pronounced in self-regions (see Supplementary Material, Figure S11). In a counterfactual analysis, even when the pathway from recurrent connections to ACW was disrupted in silico, the negative relationship between ACW and GSCORR persisted, possibly due to the influence of baseline excitation.

In summary, we replicated the empirical findings on the topographic differentiation of distinct hierarchical levels by adjusting individual nodes' basal excitation and excitatory recurrence strength, using myelin as a proxy for hierarchy. These biophysical adjustments highlight the key role of recurrent connections and basal excitatory tone in yielding the topographic-dynamic differentiation of self and non-self regions in the brain's spontaneous activity; they thus provide an explanation for the observed statistical relationships between myelin, ACW, and GSCORR in our empirical data.

3. Discussion

3.1. Overview – main findings

Our findings demonstrate that self brain regions exhibit lower myelin content, longer ACW, and decreased GSCORR compared to non-self regions (Fig. 1). These differences remain robust after controlling for preprocessing choices, unimodal-transmodal effects, and region-labeling artifacts (Supplementary Material). We further demonstrate that the distinction of self and non-self regions is not identical to the one of uni-transmodal regions; instead, we further demonstrate that the distinction of self and non-self regions also holds within the uni- and transmodal themselves respectively – this further underlines the intrinsic nature of the self-nonself distinction as one essential design feature within the brain's global topography.

At both global and regional levels, ACW negatively correlates with myelin, while GSCORR shows a positive correlation with myelin (Figure 2). These relationships persist independently of the number of regions within each class (Supplementary Material), reinforcing that they emerge from the brain's intrinsic topographic and dynamical principles. To further examine how structural hierarchy (myelin) relates to global synchronization (GSCORR), we employed a Hierarchical Bayesian Mediation Model. Counterfactual analysis confirmed that ACW mediates the myelin-GSCORR relationship in both self and non-self regions, even at the individual subject level. Additionally, GSCORR remains consistently higher in non-self regions, emphasizing the role of regularizing priors (Fig. 3). Notably, modeling ACW as a mediator improved the predictive accuracy more than a direct myelin-GSCORR relationship – this suggests that local intra-regional neural dynamics (ACW) regulate and drives inter-regional large-scale functional organization (GSCORR).

The topographic distinction of self and non-self regions is further supported by their physiological differences as investigated in computational modeling using a neural mass model, the Wilson Cown model.

After fitting and replicating the data to/in the model, we demonstrate that self regions exhibit higher levels of both recurrent excitatory connections and neural excitation compared to non-self regions. This demonstrate that the neural differences of self and non-self regions in their structural (myelin), functional (GSCORR) and dynamic (ACW) features are related to underlying physiological features, namely recurrent connections and neural excitation. Albeit tentatively, one may suggest that increased recurrent connections, together with higher neural excitation and longer autocorrelation windows, are ideally suited for the self regions' high degree of self-referential processing – the latter may then be characterized by the prolonged processing of inputs/stimuli in an echo-chamber like way.

3.2. Distinction of self and non-self regions – an intrinsic feature of the brain's global topography

We showed in our fMRI data that self and non-self regions can be distinguished on multiple grounds. Their structural features differ as they exhibit different levels of myelin, their functional connectivity is different with self regions exhibiting lower global functional connectivity (GSCORR), and their timescales also differ as they are longer in self than non-self regions (ACW). We here for the first time lower myelin and lower GSCORR and longer ACW in self regions in fMRI. This aligns with EEG findings where longer ACW in several studies during both rest and task states has been observed during self-specific stimuli.

In addition to these specifics, our findings add substantially to the present literature on self by pursuing a novel approach, a global-topographic approach. Usually, the focus in self-studies is on task-related activity to identify specific regions or networks related to self-specific stimuli as distinguished from non-self-specific ones. Additionally, a number of studies showed that the brain's spontaneous activity predicts task-related activity and self-specificity on both neural and psychological levels (Qin and Northoff, 2011; Wolff et al., 2019; Davey et al., 2016; D'Argembeau et al., 2005; Kolvoort et al., 2020; Whitfield-Gabrieli et al., 2011; Bai et al., 2016; Schneider et al., 2008).

These findings on especially the relationship of spontaneous activity and self raise the question whether the distinction of self and non-self regions is an intrinsic feature of the brain's global topography. The brain's global topography has recently been characterized by its uni-transmodal gradients. Is the distinction of self and non-self regions just a corollary of the uni-transmodal distinction or, alternatively, an intrinsic topographic feature by itself? The differences of self and non-self regions in their myelin, GSCORR and ACW clearly support the topographic distinction of these regions. Moreover, both data and model show that even within the uni- and transmodal regions themselves, we could respectively distinguish self and non-self regions. These findings speak for the second hypothesis, namely that the distinction of self and non-self regions is an intrinsic topographic feature of the brain's overall global topographic design.

Together, our structural and functional features seem to predispose the distinction of self and non-self regions already at the level of the spontaneous activity's topography; that, in turn, may drive and facilitate the task-related impact of the self on cognition and emotion as measured by, the self-prioritization effect (Northoff, 2016; Sui and Humphreys, 2015). The self prioritization effect extends to nearly all brain functions, including those mediated by non-self regions like the sensory regions. One would therefore, assume that the intrinsic topographic and dynamic balance of self and non-self regions within the brain's global uni-transmodal topography, as demonstrated here, may be key in mediating the sensory, cognitive, emotional effects of the self, that is, its domain-independent self-prioritization effect.

3.3. Distinction of self and non-self regions – higher recurrence and increased neural excitation

Our extensive statistical modeling of the empirical fMRI data suggests a specific relationship from structure over dynamics to functional connectivity. Specifically, we demonstrate that the relationship of the structural myelin to the inter-regional functional connectivity with its global synchronization (GSCORR) is mediated by the intra-regional dynamics of the timescales, e.g., the ACW. This is further supported by our modeling.

Mechanistically, neural mass modeling revealed that scaling excitatory recurrent connectivity according to myelin content first alters local temporal dynamics (ACW), which in turn influences large-scale synchronization (GSCORR). This supports a causal chain: structural hierarchy (myelin) → local temporal persistence (ACW) → global signal correlation (GSCORR). Additionally, our findings demonstrate that treating GSCORR as a mediator instead of ACW results in significantly poorer predictive performance. While our study is cross-sectional, this framework aligns with neurophysiological and computational principles, requiring future longitudinal or perturbation studies for causal validation. This trilateral relationship of myeline, GSCORR and ACW holds in both self and non-self regions. However, we observed some subtle differences in that relationship between self and non-self regions. That again suggests the intrinsic nature of the distinction of self and non-self regions within the brain's global topography.

That is further underlined by our physiological findings. After fitting the data to the model, we investigated how GSCORR and ACW in self and non-self regions are related to physiological features like recurrent connections and neural excitation. Self regions showed higher recurrent excitatory connections as well as increased neural excitation (basal excitation) compared to non-self regions. Specifically, we demonstrated that the recurrent connections are related to the ACW: higher recurrent connections go along with longer ACW with the latter thus facilitating the increased processing of the same inputs/stimuli within especially the self regions while that relationship is much weaker in non-self regions due to their lower recurrent connections and shorter ACW. Moreover, we observed that self regions exhibit higher degrees of neural excitation which, again, may facilitate the increased processing of inputs or stimuli within one region. The intra-regional nature of such input processing in the self regions is further supported by their low GSCORR, meaning that they are less impacted by the global brain's activity than the non-self regions.

In summary, together with the longer ACW and increased recurrent connections, this amounts to an increased/intense (neural excitation is high), prolonged (long ACW) and recursive (high recurrent connections) of one and the same input or stimulus within the self regions – the inputs/stimuli are processed in a recurrent way in an echo-chamber like way, an internal echo within the brain's self regions. Although tentatively, we assume that such prolonged, intense, and recursive processing leads to the assignment of higher degrees of self-referentiality of the inputs/stimuli thus entailing higher self-prioritization. Self-referential processing may then be conceived as a form of 'psychological recurrence' where the inputs/stimuli longer, intense, and repetitive processing makes more likely their association with the self. As such psychological and physiological levels of self-referential processing may be connected with each other through their sharing of one and the same feature, that is, recurrence which may thus provide their "common currency" (Northoff et al., 2025; Northoff et al., 2020) (Fig. 6).

4. Limitations

Firstly, task-based data were not included, although self-related regions were derived from task-based meta-analyses. Despite substantial evidence linking self-related processes to resting state activity, the absence of task data limits the ability to generalize findings across different cognitive states. Secondly, the study did not incorporate psychological measures or behavioral data, precluding the possibility of correlating structural and functional brain self – nonself differences with specific cognitive or emotional outcomes. Future research should integrate task-based data and psychological assessments to better understand the self/non-self distinction in brain activity and function. Additionally, incorporating behavioral measures would offer valuable insights into how these neural distinctions translate into observable behavior.

Similar to Deco et al., our approach incorporates a hierarchical model using myelin (T1w/T2w ratio), yet differs in how hierarchy is defined (Deco et al., 2021). Deco's model employs thermodynamic measures of information reversibility (Kringelbach et al., 2024), whereas our model treats hierarchy as a gradient of excitatory parameters (i.e., recurrent connections and basal excitation) progressively increasing through the cortex (Hilgetag and Goulas, 2020). This aligns with previous findings showing that intrinsic timescales (ACW) follow an organizational gradient, extending neurophysiological insights into cortical hierarchy and excitation (Manea et al., 2022).

There are potential confounders for the mediation model. Given the known developmental and aging trajectories of myelin and intrinsic timescales, future studies should examine age-related variability more explicitly (Bartzokis et al., 2010; Wu and Gollo, 2025). Here, HCP S1200 Subject dataset is homogenous for age, which decrease potential confounding effect detrimentally. Also, cortical hierarchy through myelin is estimated by T1w/T2w ratio of MRI, that can be affected by vascular density and metabolic activity, not just myelin (Sandrone et al., 2023). Thus, regional differences in ACW and GSCORR could partially reflect vascular confounds and iron (Birkl et al., 2019) rather than direct neural properties. Finally, alternative measures of functional connectivity, including metastability and effective connectivity, may provide complementary insights into the role of hierarchy in shaping large-scale brain dynamics.

5. Conclusion

Our study provides critical insights into how the intricate relationship of how structural-functional hierarchy, neural dynamics, and physiological features shape the distinction of self- and non-self regions within the global topography of the brain's spontaneous activity. We use fMRI resting state to demonstrate that self regions exhibit lower myelin content, lower global signal correlation (GSCORR), and longer autocorrelation windows (ACW) when compared to non-self regions. Hierarchical Bayesian modeling and mediation analyses show that ACW is key in mediating the impact of myelin on GSCORR in both self and non-self regions albeit in slightly distinct ways.

Next, after fitting the data to the Wilson-Cowan model, we show that the distinction of self and non-self regions does not only hold globally but also within the transmodal and unimodal regions themselves respectively. This strongly supports the assumption of the intrinsic nature of the self-non-self regions within the brain's global topography: the distinction of self and non-self is an essential or integral element of the brain's topographic design. The distinction of self and non-self regions is further supported by their physiological differences: self regions show higher recurrent connections and increased neural excitation than the non-self regions. The higher degree of recurrent connections, together with increased neural excitation and longer autocorrelation windows, are supposedly ideal to mediate these regions' self-referential processing – this may thus be seen as form of 'psychological recurrence' where one and the same input/stimulus is processed in a prolonged echo-chamber like way, an internal echo within the brain's self regions.

6. Methods

6.1. Dataset

6.1.1. Ethics approval

The data used in this study were obtained from the Human Connectome Project (HCP) database (WU-Minn Consortium; Principal Investigators: David Van Essen and Kamil Ugurbil). All procedures for data acquisition and sharing were approved by the Washington University Institutional Review Board (IRB #201,204,036), and all participants provided written informed consent. No new data were collected by the authors. All procedures were conducted in accordance with relevant institutional and national ethical guidelines and regulations.

6.1.2. Data processing

We utilized 7T MRI scans from HCP S1200 Subjects dataset. We included subjects who underwent 7T fMRI but did not have a corresponding 3T sequence, allowing us to compare results with 3T fMRI scans as independent dataset, without overlapping subjects in two groups. This cohort consisted of a total of 164 subjects with 7T data. The reader can find the relevant information about dataset in our Supplementary Material – Dataset, processing pipeline, calculation of intracortical myelin content, estimation of neural timescales and global signal correlation in the Supplementary Material. In summary, myelin content is approximated by T1/T2 ratio (Glasser and Van Essen, 2011; Sui et al., 2022) (Supplementary Material). We estimated intrinsic neural timescales (INT) using the autocorrelation function, where ACW-0 (the first lag at which the autocorrelation drops below zero) was calculated for each Glasser parcellation of BOLD time-series data, averaging across time windows (5 s) to derive a mean ACW-0 value (Supplementary Material- Global signal correlation (GSCORR) was computed as the Fisher-Z transformation of the Pearson correlation between the BOLD signal within each voxel/region and the averaged gray matter BOLD signal, producing a topographic measure of each region's correlation to the global brain activity.

It is important to note that, in the main text, we only detrended the fMRI data without applying bandpassing. However, we also present results with bandpassing and global signal regression, using a 2 × 2 repeated measures ANOVA design to compare the findings with those obtained from the HCP preprocessing pipeline (ICA-FIX) (See Supplementary Material).

6.2. Defining self and non-self regions

Regions associated with the self were identified based on a prior large-scale meta-analysis of self-related areas (Qin et al., 2020). The MNI coordinates of these regions were mapped onto the Glasser Parcellation using the AFNI software to determine their positions within the parcellation scheme. All parcellations not labeled as self-regions were classified as non-self, resulting in a total of 33 self regions and 327 non-self regions (see Fig. 1 for the regional details). Table for regions presented in Supplementary Material.

To avoid introducing a false difference between self and non-self regions, we shuffled the region labels across the whole brain while maintaining the same numbers for each class (33 for self and 327 for non-self). We performed this shuffling in two ways: (1) complete random shuffling, repeated 10 (Wolff et al., 2019) times, and (2) controlled shuffling, where self-regions were shuffled uniformly and evenly across hemispheres. For further details, see Supplementary Material – Bootstrapping and Results.

7. Statistical methods

7.1. Hypothesis testing

To compare differences among variables in self and non-self regions two distinct analysis applied. Firstly, we averaged the results of each variable (ACW-0, myelin and GSCORR) within self and non-self regions for each subjects. We compared self and non-self regions with paired *t*-test. Secondly, we averaged subjects for each region and compared self and nonself region groups with Wilcoxon rank test for MY and GSCORR but *t*-test for ACW, due to non-normal distribution of MY and GSCORR in non-self (See Fig. 1A and B).

7.2. Spatial permutation test

To explore the relationships among myelin content, GSCORR, and ACW-0, we first examined the correlations between myelin and ACW-0, as well as between myelin and GSCORR, to assess structural-functional coupling across the entire brain (global), including self and non-self regions.

Spatial permutation analysis was employed to address potential autocorrelation among these variables, as well as to consider hemispheric symmetry and location. This approach entails generating spatial autocorrelation-preserving null models for each hemisphere. Specifically, we compare the empirical two-tailed Spearman's correlation between two spatial maps (each representing a variable) with a null distribution of Spearman's correlations. This null distribution is derived by projecting one of the spatial variables onto a sphere, randomly rotating this sphere, and then re-projecting the rotated map back onto the cortical surface (Alexander-Bloch et al., 2018). In our analysis, this procedure was repeated 100,000 times to establish a comprehensive distribution. Lastly, we compared the Spearman correlation coefficient with null distribution and reported the significance that was named as p_{spin} (Significance for p_{spin} threshold is 0.05) (Alexander-Bloch et al., 2018)(See Results and Figure 2)

7.3. Hierarchical varying intercept and slope model

To investigate the mediating effect of INT (ACW) on the relationship between myelin content (MY) and global signal correlation (GSCORR), we employed Directed acyclic graphs with bayesian regression approach utilizing the 'rstan' (Stan Development Team 2025), 'cmdstanr' (Gabry et al., 2024) and 'rethinking' (McElreath, 2020) packages within R version 4.2.2. To elucidate the mediating influence of INT (ACW) on the relationship between myelin content and global signal correlation (GSCORR) across self and non-self regions, we employed Bayesian estimators to navigate the complex interactions among these variables. The choice of Bayesian estimation stems from its capacity to incorporate uncertainty and potential interactions, offering a nuanced understanding of the variables' relationships. Utilizing the rethinking and rstan packages for Bayesian estimation, we coded and sampled the models in R 4.2.2, running 4 chains with 1000 iterations each. Control results for the chains are provided in the Supplementary material.

To use whole uncertainty and variability in the data, we investigated data with hierarchical priors with varying subjects, named as Hierarchical Multi-Level Bayesian Analysis. This approach allows us to regularize our estimates because we are putting hyperpriors for priors (McElreath, 2020). The model is structured to analyze data across multiple subjects and brain regions, focusing on key variables that represent structural and functional characteristics—specifically, standardized myelin content, global signal correlation, and autocorrelation windows. Standardization ensures that the clustering results are not biased by the scale of the features, leading to more reliable and interpretable outcomes. By putting all features on the same scale, standardization helps achiving better performance and more meaningful insights from the analysis. Each subject's brain region data are categorized into 'self' or 'non-self' to facilitate targeted analyses of these attributes. This setup employs varying intercepts for each subject and condition (self vs. non-self), and parameters are estimated using priors centered around zero, reflecting no initial bias towards specific outcomes. The model's robustness is ensured by employing Cholesky factorization for correlation matrices (L_med and L_dir), which capture underlying dependencies among model parameters and thus, provide a comprehensive understanding of the brain's functional architecture.

We designed two models: a baseline model to assess the relationship between GSCORR and myelin, and an alternative model proposing ACW as a mediator in this relationship. Models' parameters are in following (std stands for *standardized*):

7.3.1. Baseline model

Data Block

n_{subj}	Total number of subjects
$n_{regions}$	Total number of regions per subject
$self_{i,j}$	Defines 1 as self, 0 as nonself for subject i and region j
$nonself_{i,j}$	Defines 1 as nonself, 0 as self for subject i and region j
$GS^{std}_{i,j}$	Standardized GSCORR for subject i and region j
$MY^{std}_{i,j}$	Standardized myelin for subject i and region j
$G_{i,j}$	Grouping variable for subject i and region j

Parameters Block

ac_1, bc_1, ac_2, bc_2	Regression coefficients for MY to GS
σ	Variance for GS
σ_{MY}	Variance for multivariate distrubition for intercept and slopes
L_{dir}	Cholesky factor correlation matrix for intercepts and slopes (4 × 4)
z^i_{dir}	Latent variables for subject i (a vector of length 4)

Transformed Parameters Block

$ac^{(i)}{}_{subjself}, bc^{(i)}{}_{subjself}, ac^{(i)}{}_{subjnonself}, bc^{(i)}{}_{subjnonself}$	Varying intercept and slopes for MY to GS for subject *i*

$$ac^{(i)}{}_{subjself} = ac1 + \left(diag(\sigma MY)\cdot L_{dir}\cdot z^{(i)}{}_{dir}\right)_1$$

$$bc^{(i)}{}_{subjself} = bc1 + \left(diag(\sigma MY)\cdot L_{dir}\cdot z^{(i)}{}_{dir}\right)_2$$

$$ac^{(i)}{}_{subjnonself} = ac2 + \left(diag(\sigma MY)\cdot L_{dir}\cdot z^{(i)}{}_{dir}\right)_3$$

$$ac^{(i)}{}_{subjnonself} = ac2 + \left(diag(\sigma MY)\cdot L_{dir}\cdot z^{(i)}{}_{dir}\right)_4$$

Model Block

$$ac_1, bc_1, ac_2, bc_2 \sim N(0, 0.5)$$

$$L_{dir} \sim LKJcorrcholesky(2)$$

$$\sigma \sim Exponential(1)$$

$$\sigma_{MY} \sim Exponential(1)$$

$$z^i_{dir} \sim N(0, 1)$$

$$\mu_{MY\{i,j\}} = ac^{G\{i,j\}}_{subjself} + bc^{G\{i,j\}}_{subjself}\cdot MY^{std}_{\{i,j\}}\cdot self_{\{i,j\}} + ac^{G\{i,j\}}_{subjnonself} + bc^{G\{i,j\}}_{subjnoself}\cdot MY^{std}_{\{i,j\}}\cdot nonself_{\{i,j\}}$$

$$GS^{std}_{\{i,j\}} \sim N\left(\mu_{MY_{\{i,j\}}}, \sigma\right)$$

Generated Quantities

$$Rho_{dir} = L_{dir}\cdot L^T_{dir}$$

Alternative Model

Data Block

n_{subj}	Total number of subjects
$n_{regions}$	Total number of regions per subject
$self_{i,j}$	Defines 1 as self, 0 as nonself for subject *i* and region *j*
$nonself_{i,j}$	Defines 1 as nonself, 0 as self for subject *i* and region *j*
$GS^{std}_{i,j}$	Standardized GSCORR for subject *i* and region *j*
$MY^{std}_{i,j}$	Standardized Myelin content for subject *i* and region *j*
$ACW^{std}_{i,j}$	Standardized Autocorrelation Window for subject *i* and region *j*
$G_{i,j}$	Grouping variable for subject *i* and region *j*

Parameters Block

$a_1, b_1, c_1, a_2, b_2, c_2$	Regression coefficients for MY → GS ← ACW
ac_1, bc_1, ac_2, bc_2	Regression coefficients for MY → ACW
$\sigma_G, \sigma, \sigma_{ACW}, \sigma_2$	Variance parameters
L_{med}, L_{dir}	Cholesky factor correlation matrices
$z^{(i)}_{med}, z^{(i)}_{dir}$	Latent variables for subject *i*

Transformed Parameters Block

$\theta^{(i)}_{subj,med,self} = \left\{a^{(i)}_{\text{subj\_self}}, b^{(i)}_{\text{subj\_self}}, c^{(i)}_{\text{subj\_self}}\right\}$	Varying self parameters for subject i
$\theta^{(i)}_{subj,med,nonself} = \left\{a^{(i)}_{\text{subj\_nonself}}, b^{(i)}_{\text{subj\_nonself}}, c^{(i)}_{\text{subj\_nonself}}\right\}$	Varying nonself parameters for subject i
$\theta^{(i)}_{subj,dir} = \left\{ac^{(i)}_{\text{subj\_self}}, bc^{(i)}_{\text{subj\_self}}, ac^{(i)}_{\text{subj\_nonself}}, bc^{(i)}_{\text{subj\_nonself}}\right\}$	Varying parameters for MY → ACW for subject i
$\theta^{(i)}_{subj,dir} = \theta^{(i)}{}_{subj,dir} + \left(diag(\sigma_{ACW})\cdot L_{dir}\cdot z^{(i)}{}_{dir}\right)$	
$\theta^{(i)}_{subj,med} = \theta^{(i)}{}_{subj,med} + \left(diag(\sigma_{G})\cdot L_{med}\cdot z^{(i)}{}_{med}\right)$	

Model Block

$$a_1, b_1, c_1, a_2, b_2, c_2 \sim N(0, 0.5)$$

$$ac_1, bc_1, ac_2, bc_2 \sim N(0, 0.5)$$

$$\sigma, \sigma_G, \sigma_{\text{ACW}}, \sigma_2 \sim \text{Exponential}(1)$$

$$L_{\text{med}} \sim \text{LKJ\_corr\_cholesky}(2)$$

$$L_{\text{dir}} \sim \text{LKJ\_corr\_cholesky}(2)$$

$$z^{(i)}_{\text{med}}, z^{(i)}_{\text{dir}} \sim N(0,1) \text{for each } i$$

$$\mu^{\text{ACW}}_{i,j} = ac^{(G_{i,j})}_{\text{subj\_self}}\cdot\text{self}_{i,j} + bc^{(G_{i,j})}_{\text{subj\_self}}\cdot\text{MY}^{\text{std}}_{i,j}\cdot\text{self}_{i,j}$$

$$+ac^{(G_{i,j})}_{\text{subj\_nonself}}\cdot\text{nonself}_{i,j} + bc^{(G_{i,j})}_{\text{subj\_nonself}}\cdot\text{MY}^{\text{std}}_{i,j}\cdot\text{nonself}_{i,j}$$

$$\text{ACW}^{\text{std}}_{i,j} \sim N\left(\mu^{\text{ACW}}_{i,j}, \sigma_2\right)$$

$$\mu_{i,j} = a^{(G_{i,j})}_{\text{subj\_self}}\cdot\text{self}_{i,j} + b^{(G_{i,j})}_{\text{subj\_self}}\cdot\text{MY}^{\text{std}}_{i,j}\cdot\text{self}_{i,j}$$

$$+c^{(G_{i,j})}_{\text{subj\_self}}\cdot\text{ACW}^{\text{std}}_{i,j}\cdot\text{self}_{i,j}$$

$$+a^{(G_{i,j})}_{\text{subj\_nonself}}\cdot\text{nonself}_{i,j} + b^{(G_{i,j})}_{\text{subj\_nonself}}\cdot\text{MY}^{\text{std}}_{i,j}\cdot\text{nonself}_{i,j}$$

$$+c^{(G_{i,j})}_{\text{subj\_nonself}}\cdot\text{ACW}^{\text{std}}_{i,j}\cdot\text{nonself}_{i,j}$$

$$\text{GS}^{\text{std}}_{i,j} \sim N\left(\mu_{i,j}, \sigma\right)$$

Generated Quantities Block

$$Rho_{\text{med}} = L_{\text{med}}L^{\top}_{\text{med}}$$

$$Rho_{\text{dir}} = L_{\text{dir}}L^{\top}_{\text{dir}}$$

8. Neural mass modelling – wilson cowan model

To simulate empirical findings in a computational setting, we employed the Wilson-Cowan neural mass model. This model effectively simulates the dynamics of firing rates across brain regions, incorporating neural populations through a system of coupled differential equations between inhibitory and excitatory populations. A detailed mathematical explanation of the model can be found in other sources (Papadopoulos et al., 2020; Cakan et al., 2023).

We implemented a dynamic model using the Python library neurolib (Cakan et al., 2023). In this model, each node consists of excitatory and inhibitory neural populations that together define their behavior. The connections between these populations are bidirectional, including excitatory-to-inhibitory (CEI) and inhibitory-to-excitatory (CIE) pathways; each population also has recurrent connections to itself to support regulatory mechanisms (Fig. 5A). The nodes are interconnected using a structural connectivity matrix derived from averaged healthy human diffusion tensor imaging (DTI) data (Rosen and Halgren, 2021), based on the HCP-MMP 1.0 atlas (Glasser et al., 2016), which includes 360 regions of interest.

The model also incorporates a functional tuning parameter that modulates the strength of the global coupling (‘K_gl’ = 2.067), thereby controlling the connectivity strength between the nodes. There are two types of excitatory input in the model: a basal excitatory intrinsic tone that is time-independent, and a time-dependent excitatory input. To simulate resting-state activity, we used a Wiener white noise process as the time-dependent (Dąbrowska et al., 2021). It has been demonstrated that excitatory recurrent connections and excitatory activity regulate local oscillatory activity, which reflects neural dynamics.

To isolate the relationship between ACW, excitatory recurrent connections, and excitation, we first conducted a single-node simulation using the Wilson-Cowan model. In this simplified scenario, a single excitatory-inhibitory population was simulated without network-level interactions. We systematically varied recurrent excitation (c_excexc, 10 to 30 in increments of 0.1) and baseline excitation fixed to 0.65 measuring the resulting ACW values. We also tested fluctuations in the firing output similar to neural firing rates. This preliminary analysis confirmed that increased recurrent excitation prolongs the ACW, consistent with previous findings on cortical hierarchy and synaptic dynamics. These results informed our full network simulation, where these parameters were topographically scaled based on myelin content as proxy for hierarchy.

For the recurrent excitatory connectivity strength (‘c_excexc’), values were inversely scaled based on myelin content, as suggested in previous studies (Wang, 2020). It has been shown that moving from sensory to association cortices leads to an increase in NMDA receptor subunits, which is negatively correlated with myelin content—a proxy for cortical hierarchy (Zhang et al., 2023). In this configuration, higher-level regions, such as transmodal self and transmodal non-self areas, exhibited greater recurrent activity and higher excitatory-inhibitory balance (EIB) ratio (Zhang et al., 2023; Larsen et al., 2022).

For basal excitatory activity for each node, we increased excitatory activity across nodes according to hierarchical organization that approximated by myelin. We used a custom function to generate a synthetic variable with a specified mean, standard deviation, and correlation to the existing myelin variable. The function created a random target variable from a normal distribution, standardized it, and then combined it with a standardized version of myelin to achieve the desired correlation. This new variable was transformed back to the original scale using the specified mean and standard deviation.

We evaluated a neural model’s dynamics by performing a multi-stage simulation process. The model is initialized from the provided trajectory using random initial conditions, and key parameters—such as recurrent excitation (c_excexc) and baseline excitation (exc_ext_baseline)—are dynamically updated using a above mentioned function to introduce predefined statistical properties. Evaluation conducts simulations in three stages: (1) a brief initial check (3 s) to ensure the model produces activity, (2) a 30-second test of the BOLD signal to verify dynamic responses, and (3) a full simulation run for the duration defined in the model parameters (Cakan et al., 2023). Post-processing includes calculating the autocorrelation window (ACW) and global signal correlation (GSCORR) for both the firing rate signals. Results are stored if the model meets criteria for valid dynamic behavior; otherwise, they are saved as invalid. The simulation systematically explores parameter space using a grid search to understand the effects of different parameters on model behavior (Fig. 5B).

We defined a parameter space using a grid search approach to systematically explore the effects of different values on the model’s behavior. The parameters included the standard deviation of recurrent

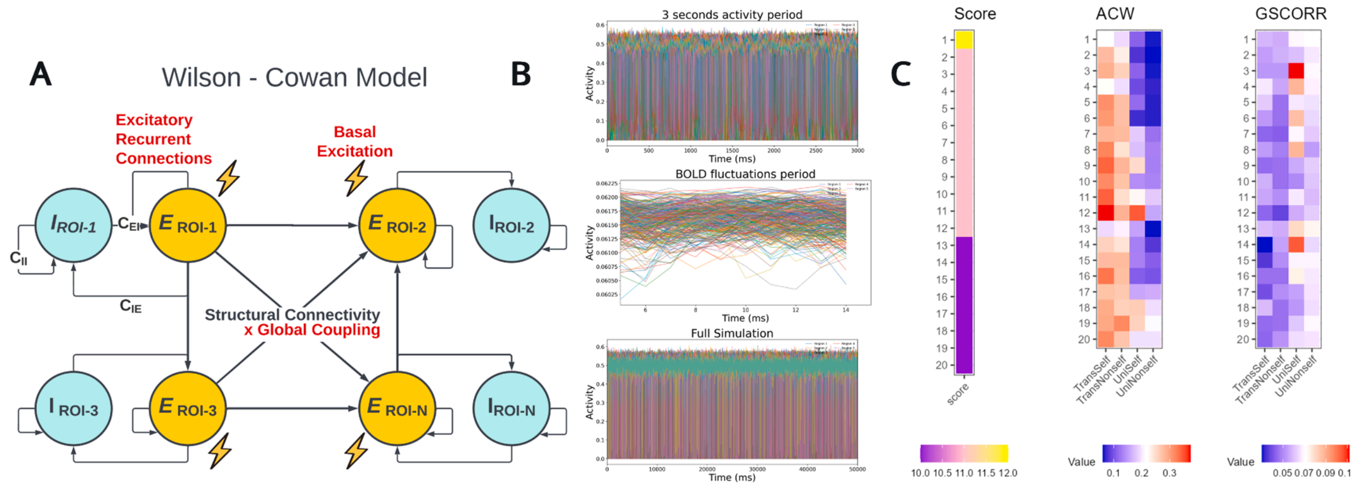

Fig. 5. *Multistep process of Neural Mass Simulation A) For neural mass modelling, Excitatory Recurrent Con*nection, Basal Excitation and Global Coupling are parameterized. B) Evaluation involves three stages: (1) a brief initial check (3 s) to ensure model activity, (2) a 30-second test of the BOLD signal to verify dynamic responses, and (3) a full simulation run for the defined duration. C) Results showing ACW and GSCORR for the best twenty parameters set.

excitation (‘sd_recur’, ranging from 1 to 5 in 5 increments) and the baseline excitation (‘sd_basal’, ranging from 0.05 to 0.2 in 5 increments), as well as their respective means (‘mean_recur’, ranging from 12 to 20 in 5 increments, and ‘mean_basal’, ranging from 0.5 to 0.8 in 4 increments). We scaled the parameters with a coefficient of −1 for recurrent connections and −1 for basal excitation to ensure that both were appropriately adjusted according to the myelin content. This grid-based parameter space allows us to comprehensively evaluate the influence of varying excitation and baseline levels on model dynamics and output measures, such as the autocorrelation window (ACW) and global signal correlation (GSCORR). Baseline excitatory mean range was selected based on evidence indicating that physiologically relevant gamma-band oscillations are observed within these limits (Papadopoulos et al., 2020).

We iterated models for 5×10 (Davey et al., 2019) milliseconds with iteration step 0.1 ms for Euler – Maruyama approximation for the coupled differential equations. The first 2000 ms was used as a burn-in period in simulation. For statistical comparison, we iterated models 60 times, and averaged regions as transmodal-self, transmodal non-self, unimodal-self and unimodal-nonself. For comparing the layers (uni/-transmodal and self and non/self) we implemented a repeated fixed-effects ANOVA model that is presented in Supplementary Material – Multiple Comparison.

To evaluate how well simulated data matched empirical observations, we employed both a pairwise comparison of category (TS, TNS, US, UNS) differences and a rank-based correlation measure. We summarized data (simulated ACW and GSCORR) for each category across parameter sets. We then compared these simulated vectors to corresponding empirical vectors using two metrics: 1- Pairwise match score: It compares the direction of the difference in the empirical data to the direction in the simulated data for pairs in categories. If the directions match, it is incremented as “match” count. Summing all matching comparisons yields a numerical “pairwise match score" that reflects how consistently the simulation reproduce the relative ordering of categories observed in the empirical dataset. 2- Spearman’s Rank Correlation: Independently, we calculated Spearman’s rho between the entire empirical vector and the corresponding simulated vector. This measure captures the degree of monotonic association, irrespective of the exact numerical differences among category means. The pairwise match score and Spearman’s rank correlation were each computed separately for the ACW and GS dimensions. These values were then combined to produce overall metrics (total match score across both traits and the average Spearman’s correlation) and used to rank the parameter settings according to how closely the simulated data reproduced observed empirical patterns.

To test the effects of both recurrent connections (`c_excexc`) and excitation (`c_ext_baseline`), we examined two additional conditions: (1) a "nontopographical" approach, where the recurrence strength of excitatory nodes was fixed (`c_excexc` = 20) along with a fixed basal excitation (`c_ext_baseline` = 0.8); and (2) a condition with basal excitation scaled as in the topographically replicated method but with fixed recurrent connections (`c_excexc` = 20). These alternatives allow us to test effect of myelin scaling, and, effect of recurrent connections, respectively.

8.1. Mixed effect model

We examined the effects of excitatory recurrent connections and global coupling using an iterative approach. First, we maintained the same topographic framework and simulation parameters, and then performed a grid search for global coupling values ranging from 0.5 to 5. Our evaluation strategy (detailed in the main text) indicated that global coupling values above 2.6 yield biologically implausible outputs. For each simulation, both ACW and GSCORR measures were computed.

To investigate the relationships between these measures and the continuous parameters (global coupling and excitatory recurrent connections), we employed mixed-effects models. In these models, the continuous parameters served as fixed predictors, while region categories (TransSelf, TransNonself, UniSelf, and UniNonself) were included, with regions modeled as random effects. An example of the model is as follows:

$$ACW_{ij} = \beta_0 + \beta_1 * Recurrent_{ij} + \beta_2 * Category_{ij} + \beta_3 * \left(Recurrent_{ij} \times Category_{ij} \right) + \beta_{0_j} + \varepsilon_{ij}$$

The output (acw) for observation i in region j, β_0 denotes the fixed intercept, β_1 is fixed effect of recurrent connections, β_2 is the fixed effect coefficent for categories, β_3 is the interaction coefficient for the product of recurrent and region categories, β_{0_j} is the random intercept for region j, ε_{ij} is the residual error for observation i in region j. We have changed that equation for two outputs (i.e. ACW and GSCORR) and two predictors (i.e. excitatory recurrent connections and global coupling). We utilized Nakagawa’s R2 for mixed models to quantify how much of the variability in the explained by the model for fixed (marginal R2) and random effects(conditional R2 – marginal R2), separately (Nakagawa and Schielzeth, 2013).

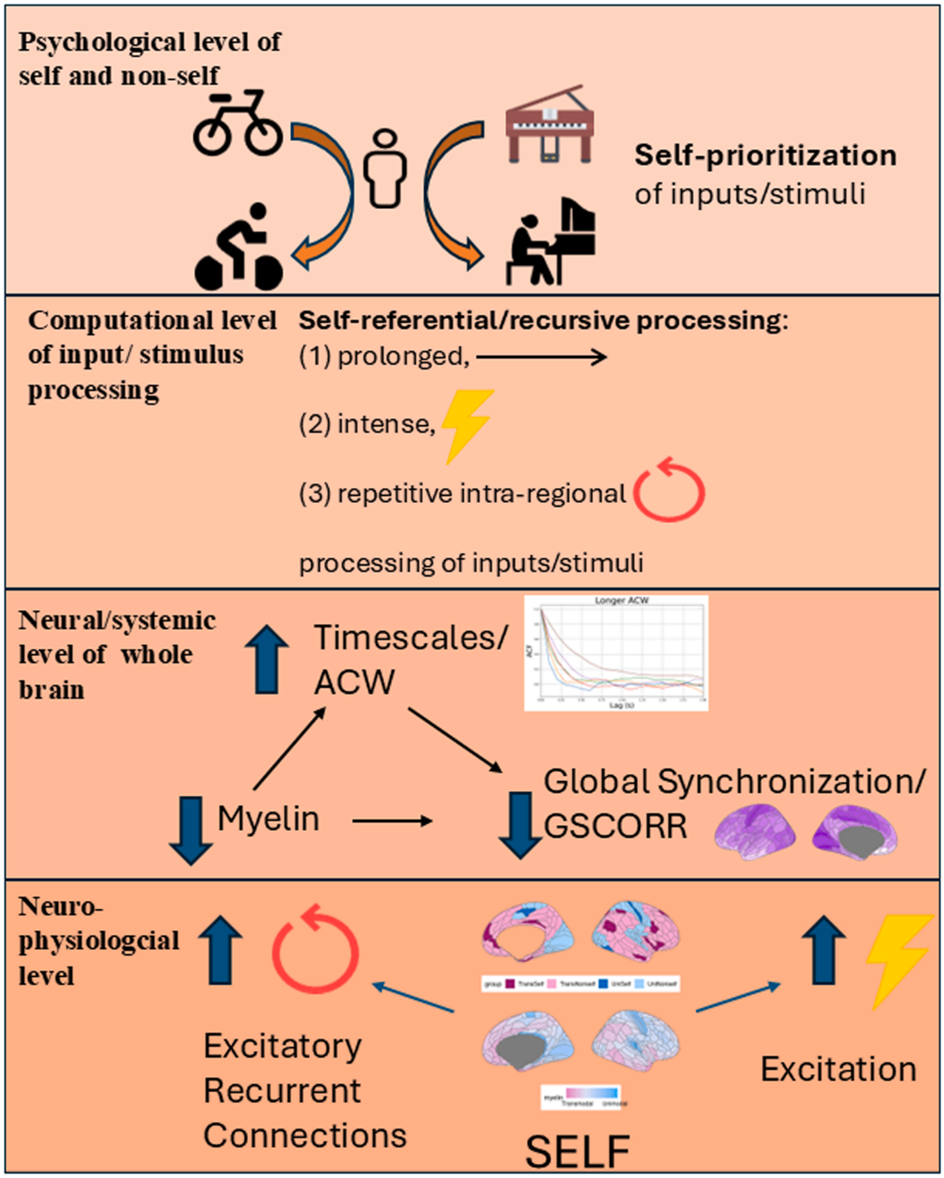

Fig. 6. Self regions demonstrate higher recurrent excitatory connections and higher basal excitation than non-self regions at neurophysiological level. Functionally, this creates a longer ACW and lower GSCORR in self regions than non-self regions. This neuronal mechanism operating like an "echo chamber" by increasing autocorrelation and thus ACW, that can reflect as self-prioritization effect psychologically.

Data and code availability

The fMRI data analyzed in this study were obtained from the Human Connectome Project (HCP) and are publicly available at https://www.humanconnectome.org/study/hcp-young-adult. The data comply with the HCP's open-access guidelines and can be accessed by registering for a data use agreement.

Processed data generated during this study, including derived autocorrelation windows (ACW) and global signal correlation (GSCORR) metrics, have been deposited in Zenodo and are publicly available at https://zenodo.org/records/14,587,447 (DOI: https://doi.org/10.5281/zenodo.14587447).

Code availability

All original analysis and modeling scripts used in this study have been deposited in GitHub and are publicly available at https://github.com/kaanka5312/MYELIN_2.git. The repository includes:

Preprocessing scripts for ACW and GSCORR computation.

Bayesian modeling scripts implemented in Stan.

Neural mass modeling scripts developed in Python.

Helper functions and topography strings, which indicate the locations of regions in the topography, are available at https://github.com/kaanka5312/NeuroMyelFC.git.

Additional Information

Any additional information required to reanalyze the data reported in this study is available from the lead contact upon request.

CRediT authorship contribution statement

Kaan Keskin: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Yasir Catal:** Writing – review & editing, Software, Conceptualization. **Angelika Wolman:** Writing – review & editing, Software, Methodology. **Mehmet Cagdas Eker:** Writing – review & editing, Supervision, Methodology. **Ali Saffet Gonul:** Writing – review & editing, Supervision, Methodology. **Georg Northoff:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.neuroimage.2025.121221.

Data availability

We have shared the processed data on Zenodo (DOI:10.5281/zenodo.14587446) and utilized the Human Connectome Project (HCP) Young Adult Open Dataset for the project.

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